COMPUTE PRICE

Methodology

SAVED PRICE SNAPSHOT
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An estimate of **whole-machine rental cost**, in USD/hour. The catalog contains **4,049 configurations** from AWS, Microsoft Azure, Google Cloud, Alibaba Cloud, Oracle Cloud and IBM Cloud. Public Linux on-demand prices use one US region per provider; discounts, taxes and separately billed storage/traffic are excluded.

THE TWO PUBLISHED MODELS

$$\text{CPU:}\quad P = H = 0.25888\,(R/16)^{0.71840}\,(C/5000)^{0.44859}$$

$$\text{GPU:}\quad P = H + 0.00171247\,nB$$

P: full VM price ($/hour). **R**: system RAM (GiB). **C**: whole-VM Geekbench 6 multi-core score. **H**: estimated CPU/RAM cost. **n**: identical full GPUs. **B**: memory bandwidth per GPU (decimal GB/s). These are fitted contributions, not provider component charges.

Data & fitting

CPU: 2,025 VMs / 326 families. Exact-VM Linux Geekbench 6.1-6.7 scores (no SME) enter least-squares fitting of log price on log RAM and log score. Missing scores are excluded.

GPU: 82 VMs / 12 families / 8 GPU models. Fits cover AWS/Azure/Google. H uses CPU-only VMs; nonnegative k minimizes whole-VM squared log error, weighting families equally. Prices below H stay included.

Fit-plot validation

Related sizes stay together: CPU validation uses five folds of provider-machine families; GPU validation holds out one family at a time, also excluding it from CPU baseline fitting.

Fit plots compare held-out predictions with actual prices on log scales; published equations use all eligible data. A second GPU test withholds whole GPU models and their host families. Error is mean absolute percentage error (MAPE), averaged over tested VMs.

Model comparisons

GPU models below use the same 82 configurations, CPU/RAM baseline and validation splits. Lower prediction error is better.

GPU contribution	Unseen family	Unseen GPU	Log $R^{2[1]}$
Bandwidth · linear (published)	22.3%	25.3%	0.935
VRAM · linear	38.0%	43.3%	0.783
Bandwidth · power law	24.5%	32.7%	0.933
VRAM · power law	39.3%	49.5%	0.817

[1] Log R^2 uses the unseen-family predictions. All four models predict every test VM.

Why bandwidth? VRAM and bandwidth correlate strongly (Pearson r = 0.845; 0.788 on logs). Adding VRAM and FP32 to the GPU fit gives zero full-sample coefficients and no better held-out error here; redundancy is sample-specific.
CPU alternative. On the same 2,025 VMs, RAM + vCPU count has 20.0% error versus 27.5% for RAM + Geekbench. vCPU speed is not standardized. The saved vCPU model tests 3,542 VMs at 21.7% error on a broader cohort with nested feature selection.

Historical reference prices

Fixed reference: base Mac mini M6, 16 GB unified memory, 20,525 Geekbench 6 CPU score (one public run) and 153 GB/s bandwidth. This estimates cloud cost at that resource level; macOS/SME scores and shared GPU memory are proxies for Linux VMs and discrete GPUs. Each dated AWS US East Linux on-demand catalog (2016-2025, plus 2026) is refitted with fixed later VM measurements, not contemporary benchmarks. Exact shapes must match RAM/vCPU specifications. Coverage changes create survivor bias; 2016 has two GPU shapes in one family. The reference extrapolates, especially early. Geekbench 6 launched in February 2023.

Limits. Model comparisons are exploratory, not independent validation of a chosen winner. Samples are not market-share weighted. Bandwidth is theoretical; daily price refreshes do not rerun benchmarks. Current fits exclude prices older than 36 hours, incompatible benchmarks and unverified units.

Sources: AWS · Azure · Google · Alibaba · Oracle · IBM · GPU specs · Geekbench 6 · AWS archives · Mac mini · M6 score

CPU benchmarks and metadata adapted from Spare Cores Inspector under CC BY-SA 4.0. Per-record sources and all earlier model results remain in the saved datasets and reports.